

Find benchmark percentages

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find 25% of 360. Copy or complete the first step.

2. Find 35% of 480.

3. Find 17% of 650.

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find 25% of 360. Copy or complete the first step.

Answer 90.

2. Find 35% of 480.

Answer 168.

3. Find 17% of 650.

Answer 110.5.

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

Answer Calculate the percentage of the original amount first, then apply that change in the context.

Find benchmark percentages

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find 25% of 360.

6. Check this result using an inverse, estimate or known fact: Find 17% of 650.

2. Find 35% of 480.

7. Prove or explain your reasoning: A £240 coat is reduced by 15%. What is the sale price?

3. Find 17% of 650.

8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.

4. Solve this independently and show every stage: Find 25% of 360.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find 35% of 480.

10. Write and solve a similar question to: Find 17% of 650.

Teacher answers and checking prompts

1. Find 25% of 360.
Answer 90.
2. Find 35% of 480.
Answer 168.
3. Find 17% of 650.
Answer 110.5.
4. Solve this independently and show every stage: Find 25% of 360.
Answer 90.
5. Use a second method or representation for: Find 35% of 480.
Answer Expected result: 168. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find 17% of 650.
Answer Expected result: 110.5. A valid check must be shown.
7. Prove or explain your reasoning: A £240 coat is reduced by 15%. What is the sale price?
Answer £204. The explanation must show why the method works.
8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.
Answer Calculate the percentage of the original amount first, then apply that change in the context.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find 17% of 650.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find benchmark percentages

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A £240 coat is reduced by 15%. What is the sale price?

6. Create a new example that tests the same idea as: Find 17% of 650.

2. Prove the answer to this question, rather than only calculating it: Find 25% of 360.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find 35% of 480.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 110.5.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A £240 coat is reduced by 15%. What is the sale price?
Answer £204. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find 25% of 360.
Answer Expected result: 90. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find 35% of 480.
Answer Expected result: 168. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 110.5.
Answer One valid reconstruction is: Find 17% of 650. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.
Answer Calculate the percentage of the original amount first, then apply that change in the context.
6. Create a new example that tests the same idea as: Find 17% of 650.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.